

RUOTIAN LIU

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Assistant Professor (Oct. 2024 - Present)

Room 335, DEI, Polytechnic University of Bari, Via Edoardo Orabona, 4, 70126 Bari BA, Italy

OVERVIEW

I have been trained in control theory and fault diagnosis for discrete event systems and hybrid systems. My previous research addressed cybersecurity issues in cyber physical systems modeled as networked discrete event systems, with a focus on the passive and active protection of secret states using integer linear programming and game theoretic approaches.

More recently, my research has expanded toward artificial intelligence based methods, particularly reinforcement learning for robotics and intelligent transportation systems. My ongoing research topics focus on deep reinforcement learning based obstacle avoidance for robotics, as well as collision avoidance on highways with on ramps.

WORK EXPERIENCE

Assistant Professor in Polytechnic University of Bari & LCA, Bari, Italy Project: IN2CCAM & CHORUS	<i>Oct. 2024 - Present</i>
Postdocal Fellow in Polytechnic University of Bari & LCA, Bari, Italy Project: eCharge4Driver & IN2CCAM	<i>Jun. 2022-Sep.2024</i>

EDUCATION BACKGROUND

Aix-Marseille University & LIS, Marseille, France Ph.D. in Automation, Laboratory of Information and System (LIS) Thesis: Contributions to event-driven control of batches Petri nets for steady state reaching Supervisors: Isabel Demongodin, Rabah Ammour, Leonardo Brenner Committee: Kamel Barkaoui, Sébastien Lahaye, Alessandro Giua, Dimitri Lefebvre, Zhiwu Li	<i>Sep.2018-May.2022</i>
Aix-Marseille University & LIS, Marseille, France M.E. in System Thesis: Consistency of diagnosability of composed labeled Petri nets via buffer places Supervisor: Isabel Demongodin, Rabah Ammour, Leonardo Brenner	<i>Sep.2017-Jun.2018</i>
Xidian University, Xi'an, China M.E. in Control Engineering, System Control and Automation Group Thesis: Consistency of diagnosability of composed labeled Petri nets via buffer places Supervisor: Zhiwu Li	<i>Sep.2016-Jul.2017</i>

RESEARCH INTEREST

Control theory, Fault diagnosis, Cyber physical system, Discrete event system, Hybrid system;
Reinforcement learning, Intelligent transportation system, Robotics.

PUBLICATIONS

j-Journal, c-Conference, t-Thesis, a-Abstract.

- [j14] **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “Deep Reinforcement Learning for Near-Optimal Control Sequence in Discrete Event Systems,” *IEEE Transactions on Automation Science and Engineering*, 2026, doi:10.1109/TASE.2026.3682500.
- [j13] Shaopeng Hu, **Ruotian Liu**, Maria Pia Fanti, Zhiwu Li, “Optimal Sensor Selection for K-Step Prognosability Verification and Enforcement of Unbounded Petri Nets,” *IEEE Transactions on Automatic Control*, 2026, doi:10.1109/TAC.2026.3683281.
- [j12] Maria Pia Fanti, **Ruotian Liu**, Shu Zhang, Zhou He, and Dimitri Lefebvre. “A Survey on Performance Evaluations in Time-dependent Petri Nets,” *Discrete Event Dynamic Systems*, 2026, doi:10.1007/s10626-026-00435-y.

- [j11] **Ruotian Liu**, Tengbo Li, Shaopeng Hu, Agostino Marcello Mangini, and Maria Pia Fanti. “Securing Networked Discrete Event Systems for Diagnosability under Attacks,” *IEEE Transactions on Automation Science and Engineering*, 2026, doi:10.1109/TASE.2026.3679186. (Accepted on March 2026).
- [j10] **Ruotian Liu**, Shaopeng Hu, Yihui Hu, Agostino Marcello Mangini, Maria Pia Fanti. “On Diagnosability Consistency of Composed Labeled Petri Nets via Buffer Places,” *IEEE Internet of Things Journal*, 2025, doi: 10.1109/JIOT.2025.3631606.
- [j9] Shu Zhang, **Ruotian Liu**, Yufeng Chen, Maria Pia Fanti, and Zhiwu Li. “Modeling and Analysis of Dual-Time Petri Nets With Application to Semiconductor Manufacturing Systems,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 23769-23783, 2025, doi: 10.1109/TASE.2025.3624696.
- [j8] Tengbo Li, Huorong Ren, **Ruotian Liu**, Maria Pia Fanti, and Zhiwu Li. “Online Opacity Verification of Networked Discrete Event Systems Modeled With Labeled Petri Nets,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 22686-22698, 2025, doi: 10.1109/TASE.2025.3614662.
- [j7] Yihui Hu, **Ruotian Liu**, Maria Pia Fanti and Zhiwu Li, “Robust fault diagnosis of networked discrete event systems using labeled Petri nets,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2025, doi: 10.1109/TSMC.2025.3573565.
- [j6] **Ruotian Liu**, Rabah Ammour, Leonardo Brenner and Isabel Demongodin. “Event-driven control of hybrid systems using Batches Petri nets: Application to high throughput manufacturing systems,” *IEEE Transactions on Automation Science and Engineering*, doi:10.1109/TASE.2025.3540744.
- [j5] Tengbo Li, Huorong Ren, **Ruotian Liu**, Maria Pia Fanti, and Zhiwu Li. “Fault diagnosis of labeled Petri nets under attacks using integer linear programming,” *IEEE Transactions on Automation Science and Engineering*, doi:10.1109/TASE.2025.3533311.
- [j4] **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “Synthesis of optimal stealthy attacks against diagnosability in labeled Petri nets,” *IEEE/CAA J. Autom. Sinica*, doi: 10.1109/JAS.2025.125156.
- [j3] **Ruotian Liu**, Yihui Hu, Agostino Marcello Mangini, Maria Pia Fanti. “K-corruption intermittent attacks for violating the codiagnosability,” *IEEE/CAA J. Autom. Sinica*, vol. 12, no. 1, pp. 1–14, Jan. 2025. doi: 10.1109/JAS.2024.124680.
- [j2] Yihui Hu, Ziyue Ma, **Ruotian Liu**, Maria Pia Fanti, Zhiwu Li. “Supervisor Synthesis for Labeled Petri Nets with Forbidden State Specification,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2024, doi:10.1109/TSMC.2024.3422925.
- [j1] Wei Duan, **Ruotian Liu**, Maria Pia Fanti, Christoforos N. Hadjicostis, Zhiwu Li. “Edit Mechanism Synthesis for Opacity Enforcement Under Uncertain Observations,” *IEEE Control Systems Letters*, vol. 7, pp. 2041-2046, 2023, doi: 10.1109/LCSYS.2023.3284340.
- [t1] **Ruotian Liu**. “Contributions to event driven control of batches Petri nets for steady state reaching,” Ph.D thesis 2021, <http://www.theses.fr/2021AIXM0656>.
- [c11] **Ruotian Liu**, Wei Duan, Shaopeng Hu, Agostino Marcello Mangini, Maria Pia Fanti. “Multiple Global Secrets Protection in Discrete Event Systems Via Integer Linear Programming,” 2026, American Control Conference.
- [c10] Gaetano Volpe, Antonio Salcuni, **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “A Hierarchical Framework for the Management of Carpooling Using Autonomous Vehicles,” 2026 IEEE Conference on Artificial Intelligence.
- [c9] Jie Ren, **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “Reinforcement Learning-Based Optimal Secret Protection in Discrete Event Systems,” 2026 IEEE Conference on Artificial Intelligence.
- [c8] **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “Optimal Attack Strategy Compromising Diagnosability of Automated Manufacturing Systems in Labeled Petri Nets,” *IEEE Conference on Control Technology and Applications*, San Diego, CA, USA, 2025, pp. 616-621, doi: 10.1109/CCTA53793.2025.11151382.
- [c7] Shu Zhang, **Ruotian Liu**, Yufeng Chen, Xuya Cong, and Maria Pia Fanti. “Modeling and analysis of time dependent Petri nets,” *IEEE 20th International Conference on Automation Science and Engineering (CASE)*, Bari, Italy, 2024, pp. 2140-2145, doi: 10.1109/CASE59546.2024.10711742.
- [c6] **Ruotian Liu**, Wei Duan, Agostino Marcello Mangini, Maria Pia Fanti. “Attack synthesis in discrete event systems under asymmetric information setting,” 17th IFAC Workshop on Discrete Event Systems, Rio de Janeiro, Brazil, Apr.29-May.1, 2024.

- [c5] **Ruotian Liu**, Wei Duan, Agostino Marcello Mangini, Maria Pia Fanti. “K-Protection of Global Secret in Discrete Event Systems Using Supervisor Control,” *2023 IEEE International Conference on Systems, Man, and Cybernetics, Honolulu, USA, 2023*, pp. 2832-2837, doi: 10.1109/SMC53992.2023.10394245.(Presenter)
- [c4] **Ruotian Liu**, Agostino Marcello Mangini, Maria Pia Fanti. “K-corruption Intermittent Attacks for Violating the Diagnosability,” *IFAC-PapersOnLine*, vol. 56, pp. 1847-1852, 2023, <https://doi.org/10.1016/j.ifacol.2023.10.1900>.
- [c3] **Ruotian Liu**, Rabah Ammour, Leonardo Brenner and Isabel Demongodin. “ON/OFF control for reaching a steady state attractive region in batches Petri nets,” *IFAC-PapersOnLine*, vol. 56, pp. 9618-9623, 2023, <https://doi.org/10.1016/j.ifacol.2023.10.267>.
- [c2] **Ruotian Liu**, Rabah Ammour, Leonardo Brenner and Isabel Demongodin. “ON/OFF Control Trajectory Computation for Steady State Reaching in Batches Petri Nets.” *Verification of Computer and Communication Systems. Lecture Notes in Computer Science*, vol 12519, 2020, https://doi.org/10.1007/978-3-030-65955-4_7.
- [c1] **Ruotian Liu**, Rabah Ammour, Leonardo Brenner and Isabel Demongodin. “Event-driven control for reaching a steady state in batches Petri nets,” *IFAC-PapersOnLine*, vol. 53, pp. 180-186, 2020, <https://doi.org/10.1016/j.ifacol.2021.04.063>.(Presenter)
- [a1] **Ruotian Liu**, Rabah Ammour, Leonardo Brenner and Isabel Demongodin. “Algorithm for controlling the transient behavior of controlled generalized batches Petri nets,” *MSR 2019 - 12ème Colloque sur la Modélisation des Systèmes Réactifs*, Nov 2019, Angers, France, Nov 2019, Angers, France. (hal-02432706) (Presenter)
- [u10] **Ruotian Liu**, Shaopeng Hu, Wei Duan, Agostino Marcello Mangini, and Maria Pia Fanti. “Deep Reinforcement Learning for Highway Traffic Management Using Proximal Policy Optimization Algorithm,” *IFAC WORLD CONGRESS 2026*. (Under Review)
- [u9] Wei Duan, Shaopeng Hu, Zhou He, **Ruotian Liu**, and Maria Pia Fanti. “Enforcing Opacity with Publicly Known Edit Functions under Incomparable Observations,” *Nonlinear Analysis: Hybrid Systems*. (2nd Review)
- [u8] Shaopeng Hu, **Ruotian Liu**, Wei Duan, Zhou He, and Maria Pia Fanti. “Critical Observability Verification in Discrete Event Systems Under Coordinated Sensor Attacks,” *Nonlinear Analysis: Hybrid Systems*. (1st Review)
- [u5] Shu Zhang, **Ruotian Liu**, Yufeng Chen, Maria Pia Fanti, Ding Liu, and Boyu Dong. “Temporal Liveness Enforcement via Parameter Tuning in Dual-Time Petri Nets,” *IEEE Transactions on Automation Science and Engineering*. (2nd Review).
- [u4] Shaopeng Hu, **Ruotian Liu**, Maria Pia Fanti, Zhiwu Li, “Optimal Sensor Attacks Undermining Prognosability and K-Prognosability in Labeled Petri Nets,” *IEEE/CAA J. Autom. Sinica*. (2nd Review)
- [u2] Wei Duan, **Ruotian Liu**, Maria Pia Fanti, Christoforos N. Hadjicostis, Zhiwu Li. “Opacity Enforcement by Edit Functions Under Incomparable Observations,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. (2nd Review)
- [u1] Wei Duan, Shaopeng Hu, **Ruotian Liu**, and Maria Pia Fanti. “A Trade-off Analysis of Multi-objective Privacy Enforcement Strategies for Discrete Event Systems,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. (2nd Review)

PROFESSIONAL ACTIVITIES

Teaching Activities

- Artificial Intelligence for Automation, Polytechnic University of Bari (60*2 hours) *Feb.2025 - Jun.2025(2026)*
I was/am fully responsible for teaching this course, covering various models for real-world automation systems, including Petri nets, automata, and Markov chains. The course also introduced artificial intelligence, with a focus on reinforcement learning from both theoretical and applied perspectives. Students were assigned projects under the theme of “automation with reinforcement learning”, such as robotic collision avoidance and scheduling in timed Petri nets.
- Control and Security of Cyber Physical Systems, Polytechnic University of Bari (10*2 hours) *Mar.2025(2026)*
This course explores automation and cyber-physical systems (CPS) from both control and security perspectives, covering formal modeling, Petri nets, and automata, with an emphasis on integrating theoretical foundations and real-world applications.

- Artificial Intelligence for Automation, Polytechnic University of Bari (30 hours) *Feb.2024-Jun.2024*
I delivered lectures on the components of Petri nets and the principles of deep reinforcement learning. Additionally, I assisted students in managing their final projects, which included topics such as collision avoidance of a planar arm via deep reinforcement learning and scheduling problems in time Petri nets.

Supervision Experience

Ph.D Students: Tengbo Li, Shu Zhang, Jie Ren, Antonio Salcuni, Shaopeng Hu.

Master Students: All students in AI for automation.

Project Experience (Horizon Europe)

Horizon Europe funded projects in intelligent transportation systems

- CHORUS: Work Package Manager and leader. Responsible for leading one work package and actively contributing to other project work packages.
- eCharge4Drive: Project participant. Involved in technical activities related to intelligent transportation systems.
- IN2CCAM: Project participant. Contributed to research and development tasks within the project.

Guest Associate Editor of

IEEE Transactions on Automation Science and Engineering

Associate Editor of

- IEEE SMC 2026
- IEEE CASE 2026

Reviewer of

- IEEE Transactions on Automatic Control
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE Transactions on Intelligent Vehicles
- Automatica
- Information Sciences
- IEEE Control System Letters
- IFAC and IEEE International Conferences
- Nonlinear Analysis Hybrid Systems

INTERNATIONAL COOPERATION

Visiting

Xidian University, School of Electro-Mechanical Engineering, Xi'an, China

2024.5-2024.6

Xi'an University of Posts and Telecommunications

Cooperation

I have collaborated with researchers from various universities, including Xidian University (Wei Duan, Ziyue Ma, Zhiwu Li), University of Cyprus (Christoforos N. Hadjicostis), Xi'an University of Posts and Telecommunications (Yihui Hu), Aix-Marseille University (Rabah Ammour, Leonardo Brenner, Isabel Demongodin), among others.

PROFESSIONAL SKILLS

Professional skills

- Discrete event system tools: PIPE, Simuleau
- Matlab, C++, Python
- Traffic simulator: SUMO

Languages

- Chinese: Native
- English: Fluent
- French: Basic
- Italian: Basic